

Question			Marking details					Marks available															
								AO1	AO2	AO3	Total	Maths	Prac										
7	(a)	(i) (ii) (iii)	The same (1) Increases (1) Increases (1)					1 1		1	3												
	(b)	(i)	<table><tr><td>Initial speed (km/h)</td><td>Initial speed (m/s)</td><td>Thinking distance (m)</td><td>Braking distance (m)</td><td>Stopping distance (m)</td></tr><tr><td>60</td><td>16.8 (1) Accept 17</td><td>25 (1)</td><td>20 (1)</td><td>45 (1)</td></tr></table>					Initial speed (km/h)	Initial speed (m/s)	Thinking distance (m)	Braking distance (m)	Stopping distance (m)	60	16.8 (1) Accept 17	25 (1)	20 (1)	45 (1)			4	4	4	
Initial speed (km/h)	Initial speed (m/s)	Thinking distance (m)	Braking distance (m)	Stopping distance (m)																			
60	16.8 (1) Accept 17	25 (1)	20 (1)	45 (1)																			
		(ii)	Substitution into: $s = \frac{d}{t}$ ecf (1) Manipulation i.e. $t = \frac{25}{16.8}$ (1) Time = 1.488 [s] or 1.49 [s] or 1.5 [s] (1) N.B. Use of 17 m/s → 1.47 [s]					1	1 1		3	2											
		(iii)	$v = 0$ (1) Substitution: $= 16.8^2$ ecf + $(2 \times a \times 20)$ ecf (1) $a = [-]7.056$ [m/s ²] (1) N.B. Use of $u = 17$ m/s → $[-] 7.225$ [m/s ²] accept correctly rounded answers					1	1 1		3	3											
	(c)	(i)	Seat belts produce an <u>unbalanced force</u> on the occupants (1) which prevents them continuing moving forward [at a constant speed] / so the occupant remains in their seat (1) Award 1 mark for N1L correctly stated if no other creditworthy response provided					2			2												

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
		(ii)	Car produces a force on the obstacle (1) producing an <u>equal and opposite</u> force on the car (1) Award 1 mark for N3L correctly stated if no other creditworthy response provided	2			2		
		(iii)	Crumple zone increases {time / distance} of the collision OR increases the time for the change in momentum (1) Reducing the {deceleration / force} on the occupant / car (1) Award 1 mark for N2L correctly stated if no other creditworthy response provided	2			2		
	(d)		{Braking / stopping} distance would be reduced [to less than 35 m] (1) Decrease in collisions / accidents (1) Reduced severity of collisions (1) so agree To award full marks a conclusion must be present			3	3		
			Question 7 total	10	4	8	22	9	0